

TASK 2: LIBRARY DATABASE (library)

Create Database: use library

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.8.3
Microsoft Windows [Version 10.0.26200.8328]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Kanagaraj>mongosh
Current Mongosh Log ID: 6a008e800e1e7a2788abc113
Connecting to:      mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.8.3
Using MongoDB:      8.2.7
Using Mongosh:      2.8.3

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
  The server generated these startup warnings when booting
  2026-05-09T10:51:38.450+05:30: Access control is not enabled for the database. Read and write access
  to data and configuration is unrestricted
-----

test> use library
switched to db library
library>

library> |
```

2. Create Collection: books

```
mongosh mongodb://127.0.0.1:27017/ x + v
Microsoft Windows [Version 10.0.26200.8328]
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C:\Users\Kanagaraj>mongosh
Current Mongosh Log ID: 6a008e800e1e7a2788abc113
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-----
  The server generated these startup warnings when booting
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  to data and configuration is unrestricted
-----

test> use library
switched to db library
library>

library> db.createCollection("books")
{ ok: 1 }
library> |
```

3. Insert Books (one and many)

```
mongosh mongodb://127.0.0.1:27017/ x + v
C:\Users\Kanagaraj>mongosh
Current Mongosh Log ID: 6a008e800e1e7a2788abc113
Connecting to:   mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.8.3
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-----
The server generated these startup warnings when booting
2026-05-09T10:51:38.450+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----

test> use library
switched to db library
library>

library> db.createCollection("books")
{ ok: 1 }
library> db.books.insertOne({ title: "The Alchemist", author: "Paulo Coelho", year: 1988, price: 250 })
{
  acknowledged: true,
  insertedId: ObjectId('6a008fcd0e1e7a2788abc114')
}
library> db.books.insertMany([
|   { title: "Wings of Fire", author: "A.P.J Abdul Kalam", year: 1999, price: 300 },
|   { title: "Harry Potter", author: "J.K. Rowling", year: 2001, price: 500 },
|   { title: "Atomic Habits", author: "James Clear", year: 2018, price: 450 }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a0090080e1e7a2788abc115'),
    '1': ObjectId('6a0090080e1e7a2788abc116'),
    '2': ObjectId('6a0090080e1e7a2788abc117')
  }
}
library> |
```

4. Read Operations

```
mongosh mongodb://127.0.0.1:27020 > use library
library> db.books.find()
[
  {
    _id: ObjectId('6a008fcd0e1e7a2788abc114'),
    title: 'The Alchemist',
    author: 'Paulo Coelho',
    year: 1988,
    price: 250
  },
  {
    _id: ObjectId('6a0090080e1e7a2788abc115'),
    title: 'Wings of Fire',
    author: 'A.P.J Abdul Kalam',
    year: 1999,
    price: 300
  },
  {
    _id: ObjectId('6a0090080e1e7a2788abc116'),
    title: 'Harry Potter',
    author: 'J.K. Rowling',
    year: 2001,
    price: 500
  },
  {
    _id: ObjectId('6a0090080e1e7a2788abc117'),
    title: 'Atomic Habits',
    author: 'James Clear',
    year: 2018,
    price: 450
  }
]
library> |
```

5. Update Operations

```
mongosh mongodb://127.0.0.1 x + v
{
  _id: ObjectId('6a0090080e1e7a2788abc115'),
  title: 'Wings of Fire',
  author: 'A.P.J Abdul Kalam',
  year: 1999,
  price: 300
},
{
  _id: ObjectId('6a0090080e1e7a2788abc116'),
  title: 'Harry Potter',
  author: 'J.K. Rowling',
  year: 2001,
  price: 500
},
{
  _id: ObjectId('6a0090080e1e7a2788abc117'),
  title: 'Atomic Habits',
  author: 'James Clear',
  year: 2018,
  price: 450
}
]
library> db.books.updateOne({ title: "The Alchemist" }, { $set: { price: 280 } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
library> |
```

6. Delete Operations

```
mongosh mongodb://127.0.0.1:27017/127.0.0.1 x + v
author: 'A.P.J Abdul Kalam',
year: 1999,
price: 300
},
{
  _id: ObjectId('6a0090080e1e7a2788abc116'),
  title: 'Harry Potter',
  author: 'J.K. Rowling',
  year: 2001,
  price: 500
},
{
  _id: ObjectId('6a0090080e1e7a2788abc117'),
  title: 'Atomic Habits',
  author: 'James Clear',
  year: 2018,
  price: 450
}
]
library> db.books.updateOne({ title: "The Alchemist" }, { $set: { price: 280 } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
library> db.books.deleteOne({ title: "Harry Potter" })
{ acknowledged: true, deletedCount: 1 }
library> |
```

```
modifiedCount: 1,
upsertedCount: 0
}
library> db.books.deleteOne({ title: "Harry Potter" })
{ acknowledged: true, deletedCount: 1 }
library> db.books.find({ price: { $gte: 250, $lte: 450 } })
[
  {
    _id: ObjectId('6a008fcd0e1e7a2788abc114'),
    title: 'The Alchemist',
    author: 'Paulo Coelho',
    year: 1988,
    price: 280
  },
  {
    _id: ObjectId('6a0090080e1e7a2788abc115'),
    title: 'Wings of Fire',
    author: 'A.P.J Abdul Kalam',
    year: 1999,
    price: 300
  },
  {
    _id: ObjectId('6a0090080e1e7a2788abc117'),
    title: 'Atomic Habits',
    author: 'James Clear',
    year: 2018,
    price: 450
  }
]
library> |
```